

Liquid and solid layers in a thermal deep learning machine

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We study generalization in overparameterized deep neural networks using a thermal deep learning machine with a well-defined Hamiltonian. We find that learning dynamics organize the network into a solid–liquid–solid structure, where boundary layers are constrained while central layers remain flexible. These layers exhibit distinct dynamical behaviors, including timescale separation and non-equilibrium aging. We show that the solution space is structureless in liquid layers but hierarchically organized in solid layers, indicating a direct link between dynamics and landscape geometry. These results suggest a physical perspective for generalization, arising from the interplay between flexible representations and boundary-induced constraints.

I. INTRODUCTION

In deep learning, a fundamental problem is to understand how the mapping between the input data $\mathbf{S}_0$ and the output data $\mathbf{S}_L$ is established through intermediate hidden layers of neurons, denoted by *internal representations* $\mathbf{S}_l$ [1], where $l = 1 \dots L - 1$ and L is the number of layers in the DNN. Noticeable progress has been made within the framework of statistical mechanics, based on the intrinsic connection between neural networks and spin glass problems [2–17]. The statistical-mechanics theory of supervised learning of single perceptrons by Gardner [6, 7] has been extended to multi-layer perceptrons (MLPs) by Yoshino [14, 15]. Such a theory considers ensembles of machines specified by realizations of synaptic weights (connection weights between neurons) $\mathbf{J}_l$ as well as the neurons (internal representations) [1] in the hidden layers $\mathbf{S}_l$. The phase space volume of the machines, called *Gardner's volume*, is the number of machines that meet all the constraints imposed by the training data. Such constraints are specified by the boundary conditions: $\mathbf{S}_0$ and $\mathbf{S}_L$ are quenched boundaries that may be (i) generated randomly (random scenario: a random constraint satisfaction problem) or (ii) provided through a randomly generated teacher-machine (teacher-student scenario: a statistical inference problem). The replica theory for the MLP finds that typical machines in Gardner's volume exhibit a solid–liquid–solid structure with the hidden layers near the boundaries in the solid states and the middle layers in the liquid states (see Fig. 1(a)). The solid state corresponds to a glass in the (i) random scenario, and a crystal associated with the teacher ma-

chine in the (ii) teacher-student scenario. Such a structure could also balance representation and generalization, via respectively the vast microscopic states in liquid layers and the strong correlations to the input/output boundaries in solid layers.

An interesting implication of the theory, which is based on the internal representation, is that one can replace the conventional machine learning algorithms based on back-propagation by physically more sensible ones, called *thermal deep learning machines* (TDLMs). The conventional algorithms attempt to minimize a loss function that depends on the actual labels $\mathbf{S}_L$, and the predicted outputs $\mathbf{S}'_L$ generated by a sequence of highly convoluted linear and non-linear operations on $\mathbf{S}_0$. In particular, such a conventional loss function does not explicitly depend on the internal representation. In contrast, the TDLM has a well-defined Hamiltonian consisting of local interactions among the internal representation $\{\mathbf{S}_l\}$ and $\{\mathbf{J}_l\}$, with the training data imposed as fixed boundary conditions $\mathbf{S}_0$ and $\mathbf{S}_L$. With the given Hamiltonian, the TDLM can be simulated at any temperature. Within the framework of TDLMs, the learning problem can be conveniently tackled using theoretical and numerical tools developed in statistical physics.

The solid layers in the MLP arise from boundary effects. In physics, there are numerous examples of boundary-induced phenomena, including liquid-gas wetting [18], melting dynamics [19, 20], the point-to-set correlation length in supercooled liquids [21–24], and vapor-deposited ultrastable glasses [25, 26]. Deep learning provides another interesting playground where breaking of the translational symmetry imposed by the boundaries can bring non-trivial consequences.

Although the picture provided by the replica theory for MLP is interesting, the current theory considers completely random, structureless input data [14, 15], which is far from reality. Also, as a static theory, the replica

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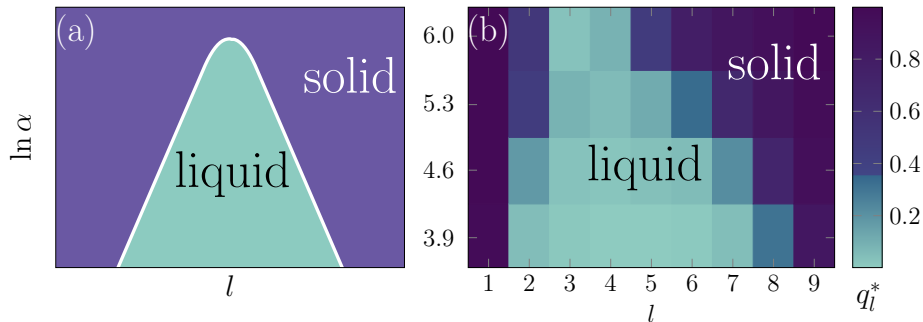


FIG. 1. **Liquid-solid phase diagram of deep learning.** (a) Schematic phase diagram for the state of network parameters in the l -th layer with a given capacity ratio $\alpha = M/N$, drawn according to the theoretical prediction Eq. (1) in the $N \rightarrow \infty$ limit [14, 15]. (b) Numerical phase diagram obtained by MD learning simulations, color coded according to the spin overlap between replicas, $q_l^* \equiv q_l(t = t^* = 5 \times 10^5, t_w = 5 \times 10^4)$. Here $L = 10, M = 2000$ are fixed, while N is varied. The boundary between liquid (green) and solid (blue) phases is estimated at $q_l^* \approx 1/e$. The numerical diagram in panel (b) is obtained at fixed $M = 2000$ by varying N .

theory does not address training dynamics. To examine the theoretical picture in deep learning of real-world data, one has to perform numerical experiments. This is the motivation of the current study.

Our simulations reveal the coexistence of liquid and solid layers in learning dynamics, summarized by a numerical phase diagram (see Fig. 1(b)). For MNIST benchmark data, the TDLM trained by a GPU-accelerated molecular dynamics (MD) algorithm has good generalization with a test accuracy up to $\sim 98\%$. The training procedure contains three dynamical regimes, with strong aging effects appearing in the intermediate time regime. Our analyses based on time-correlation functions and replica overlap order parameters show distinct dynamic and thermodynamic features of the network layers in the center and near the boundaries. The states in the central layers are liquid-like, characterized by rapid relaxation dynamics and nearly zero overlaps between differently trained machines. Meanwhile, the states in the near-boundary layers are solid-like, characterized by slow dynamics and a hierarchical organization of overlap order parameters.

The central observation of this work is that learning dynamics organizes network layers into a solid-liquid-solid structure, which simultaneously explains representation, generalization, and landscape geometry. To demonstrate this, we analyze training dynamics, generalization performance, layer-resolved behavior, and the structure of the design (solution) space. These results suggest a unified dynamical and structural framework for understanding deep learning beyond conventional optimization-based descriptions.

Recent statistical-mechanical studies have revealed complementary forms of organization in composite neural architectures. In multitasking Hebbian networks, increasing dilution drives a crossover from hierarchical retrieval, where patterns are represented with different strengths, to parallel retrieval of multiple patterns

[27]. Layered assemblies of interacting Hebbian networks display emergent pattern-disentanglement abilities that are absent in individual networks [28]. On the landscape side, replica-symmetry-breaking analyses of dense Hebbian networks have shown how hierarchical glassy states stabilize spin-glass phases, while generalized de Almeida-Thouless criteria identify the instability of replica-symmetric solutions [29, 30]. These works provide a broad context for understanding how multi-layer organization, sparsity, and complex landscape geometry can jointly shape neural-network behavior. The present study complements them by resolving the non-equilibrium, layer-dependent coexistence of liquid-like and solid-like sectors in a boundary-constrained TDLM.

II. MODELS AND METHODS

The task is to classify handwritten digits taken from the MNIST database [31, 32], with M training images and 400 test images. The TDLM employs the internal representation so that its Hamiltonian involves $\{\mathbf{S}_l\}$ and $\{\mathbf{J}_l\}$ as free dynamical variables. The canonical ensemble of the model becomes identical to the ensemble of fully connected MLPs in the zero temperature limit. We consider a rectangular network with width (the number of perceptrons in a hidden layer) N and depth (the number of layers) L . The *storage ratio* $\alpha = M/N$ quantifies the average degree of constraints per free variable in a layer. We emphasize that α is introduced here as an intensive control parameter at fixed width. It does not imply a linear bound on the storage capacity: in multilayer perceptrons, replica theory predicts that the critical capacity α_c grows exponentially with depth, $\ln \alpha_c \propto L$ [14], consistent with the known exponential expressivity of deep networks [33]. The training is accomplished by relaxing the total energy E (the expectation value of the Hamiltonian) using MD simulations at a low temperature $T = 10^{-5}$,

starting from random configurations.

This model was studied theoretically by Yoshino in Refs. [14, 15] in the zero temperature limit under the assumption that input data are completely random, and the correlations caused by interaction loops can be neglected (which is justified by considering the dense limit $N \gg c \gg 1$ with c being the connectivity of the inter-layer coupling). In the joint large- M , large- N limit for unstructured random data, the key prediction is that the *penetration depth* ξ_s of solid layers (either glass or crystal) near the boundaries linearly depends on $\ln \alpha$,

$$\xi_s \sim \ln \alpha, \quad (1)$$

according to which a schematic phase diagram is plotted in Fig. 1(a). Below we provide detailed descriptions of the model and methods.

A. Model definition: thermal deep learning machine

The thermal machine is defined on the fully connected DNN as shown in Fig. 2(a), where $S_{l,i}$ and $J_{l,i,j}$ are real numbers in $(-1, 1)$. The spins on the boundaries $\mathbf{S}_0$ and $\mathbf{S}_L$ are fixed by the input/output data respectively, while $\mathbf{S}_l$ in other layers ($0 < l < L$) and all $\mathbf{J}_l$ are free variables. The input layer, each of the $L - 1$ hidden layers, and the output layer consist of N_0 , N , and N_L neurons, respectively. Each neuron in the l -th layer is linked to every neuron in the $(l + 1)$ -th layer. In this study we set $N_0 = 784$, $N_L = 2$, $L = 10$, and a variable N (unless otherwise specified).

The energy of each layer is defined as:

$$\mathcal{E}_l(\{\mathbf{S}_{l-1}\}, \{\mathbf{S}_l\}, \{\mathbf{J}_l\}) = \sqrt{\sum_{\mu=1}^M \sum_{i=1}^{N_l} \left[S_{l,i}^\mu - \tanh \left(\sum_{j=1}^{N_{l-1}} J_{l,i,j} S_{l-1,j}^\mu \right) \right]^2}. \quad (2)$$

The Hamiltonian of the whole system is:

$$\mathcal{H}(\{\mathbf{S}_l\}, \{\mathbf{J}_l\} | \mathbf{S}_0, \mathbf{S}_L) = \sum_{l=1}^L \mathcal{E}_l(\{\mathbf{S}_{l-1}\}, \{\mathbf{S}_l\}, \{\mathbf{J}_l\}). \quad (3)$$

For a training task of M images, the boundary conditions are fixed by inputs and outputs,

$$\mathbf{S}_0 \equiv \{S_{0,1}^\mu, S_{0,2}^\mu, \dots, S_{0,i}^\mu, \dots, S_{0,N_0}^\mu\}, \quad (4)$$

$$\mathbf{S}_L \equiv \{S_{L,1}^\mu, S_{L,2}^\mu, \dots, S_{L,i}^\mu, \dots, S_{L,N_L}^\mu\}. \quad (5)$$

where $\mu = 1, \dots, M$. The free (dynamical) variables are the spin and weight parameters,

$$\mathbf{S}_l \equiv \{S_{l,1}^\mu, S_{l,2}^\mu, \dots, S_{l,i}^\mu, \dots, S_{l,N_l}^\mu\}, \quad (6)$$

and

$$\mathbf{J}_l \equiv \{J_{l,1,1}, \dots, J_{l,i,j}, \dots, J_{l,N_l,N_{l-1}}\}, \quad (7)$$

where $l = 1, \dots, L - 1$ and $\mu = 1, \dots, M$. The spin and weight parameters have $\sim LMN$ and $\sim LN^2$ total variables respectively.

The $\mu = 1, \dots, M$ training images impose M boundary conditions, or constraints, to the problem. We require that $\{\mathbf{J}_l\}$ are independent of the constraint μ , while $\{\mathbf{S}_l\}$ can depend on μ . Here, by setting $\{\mathbf{S}_l\}$ as free variables, we allow some perceptron constraints to remain unsatisfied, at the cost of a nonzero energy. When the total potential energy is zero, all constraints are satisfied, and the output matches the image label perfectly.

The corresponding statistical mechanical ensemble is characterized by a partition function,

$$Z(\mathbf{S}_0, \mathbf{S}_L) = \sum_{\{\mathbf{S}_l\}, \{\mathbf{J}_l\}} e^{-\beta \mathcal{H}(\{\mathbf{S}_l\}, \{\mathbf{J}_l\} | \mathbf{S}_0, \mathbf{S}_L)}, \quad (8)$$

where $\beta = 1/(k_B T)$ is the inverse temperature. From Eq. (8), one can see that the boundary conditions $(\mathbf{S}_0, \mathbf{S}_L)$ play the role of quenched disorder in this model.

B. Training algorithm

To train the thermal machine, we perform classical MD simulations following Langevin dynamics, at a fixed low temperature $k_B T = 10^{-5}$ (unless otherwise specified), where the Boltzmann constant $k_B = 1$. The spin and weight variables $\{\mathbf{S}_l\}$ and $\{\mathbf{J}_l\}$ are then considered as positions $\{r_i\}$ of atoms, updated by the formulas,

$$v_i(t + \Delta t/2) = v_i(t - \Delta t/2) + F_i(t) \frac{\Delta t}{m_i}, \quad (9)$$

and

$$r_i(t + \Delta t) = r_i(t) + v_i(t + \Delta t/2) \Delta t, \quad (10)$$

where v_i is the velocity, the mass $m_i = 1$ for all i , and the time step $\Delta t = 10^{-2}$. The time unit in the MD simulation is arbitrary but consistent across all measurements, allowing meaningful comparison of relative time scales. The force is given by

$$F_i(t) = -\frac{dE(r_1, r_2, \dots; t)}{dr_i} - \eta v_i(t - \Delta t/2) + \sqrt{\eta(2 - \eta)k_B T} \delta_i, \quad (11)$$

where $E(r_1, r_2, \dots; t)$ is the total energy computed according to Eq. (3) at time t , $\eta = 10^{-3}$ is the friction coefficient, and δ_i is a random variable with the standard normal distribution with a zero mean and a unit standard deviation. Because spin and weight variables are restricted in the range $(-1, 1)$, we apply reflective boundary conditions (elastic hard walls) to the simulation box. The program is accelerated using the GPU-based PyTorch software, which performs differentiation (MD force calculations) automatically and efficiently [34].

The initial conditions at $t = 0$ are random configurations of $\{\mathbf{S}_l\}$ and $\{\mathbf{J}_l\}$ drawn from uniform distributions

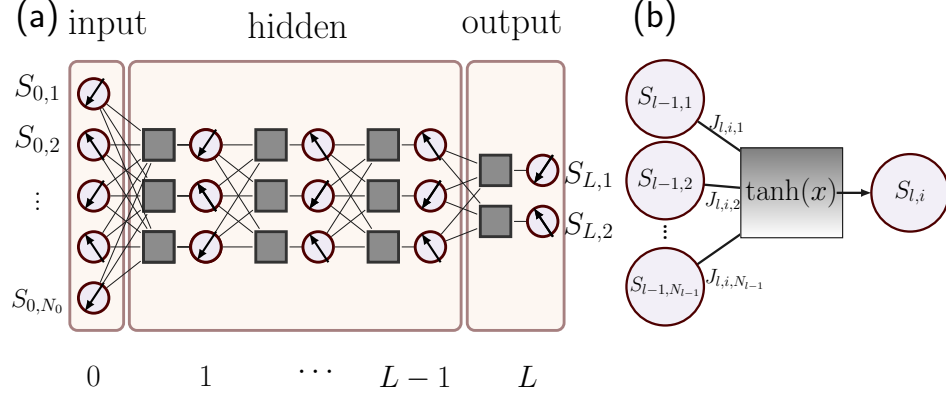


FIG. 2. **Illustration of the TDLM.** (a) The structure of the fully connected network. (b) The basic unit is similar to a single-layer feed-forward perceptron, but S_l and J_l are allowed to change dynamically.

in $(-1, 1)$. Each independent initial condition is called a sample. The dynamical data presented in this work are averaged over N_s samples. For a given sample, N_r replicas are created at a waiting time t_w . These replicas share the same particle positions at t_w , but are assigned different velocities at t_w drawn from the Maxwell-Boltzmann distribution, thus following different dynamical trajectories after t_w .

C. Feed-forward neural networks for testing

We test the $\{\mathbf{J}_l\}$ configurations learned by the TDLM in a standard feed-forward DNN. The basic unit is a single-layer perceptron,

$$S_{l,i} = \text{sgn} \left(\frac{1}{\sqrt{N_{l-1}}} \sum_{j=1}^{N_{l-1}} J_{l,i,j} S_{l-1,j} \right), \quad (12)$$

where N_{l-1} is the number of neurons in the $(l-1)$ -th layer. Each neuron is either active ($S_{l,i} = 1$) or passive ($S_{l,i} = -1$), and the weights $J_{l,i,j} \in (-1, 1)$ are copied from the TDLM. Note that $\{\mathbf{J}_l(t)\}$ evolves with time during the MD training, which allows us to test the time-dependent test accuracy $A(t)$ by copying the instantaneous $\{\mathbf{J}_l(t)\}$ to the test DNN.

D. Correlation functions

We define two types of time correlation functions. The first type is the *autocorrelation functions* for $\mathbf{S}_l$ and $\mathbf{J}_l$ in

the l -th layer, defined as,

$$c_l(t, t_w) = \frac{1}{N_l M} \sum_{i=1}^{N_l} \sum_{\mu=1}^M S_{l,i}^\mu(t + t_w) S_{l,i}^\mu(t_w), \quad (13)$$

$$C_l(t, t_w) = \frac{1}{N_l N_{l-1}} \sum_{i=1}^{N_l} \sum_{j=1}^{N_{l-1}} J_{l,i,j}(t + t_w) J_{l,i,j}(t_w), \quad (14)$$

where t_w is the waiting time, t the time elapsed after t_w , and $t + t_w$ is the total elapsed time.

The second type is the *replica-correlation functions* defined as,

$$q_l^{ab}(t, t_w) = \frac{1}{N_l M} \sum_{i=1}^{N_l} \sum_{\mu=1}^M S_{l,i}^{\mu,a}(t + t_w) S_{l,i}^{\mu,b}(t + t_w), \quad (15)$$

$$Q_l^{ab}(t, t_w) = \frac{1}{N_l N_{l-1}} \sum_{i=1}^{N_l} \sum_{j=1}^{N_{l-1}} J_{l,i,j}^a(t + t_w) J_{l,i,j}^b(t + t_w), \quad (16)$$

where $a, b = 1, \dots, N_s \times N_r$ label the replicas. For each of the N_s independently initialized samples, N_r replicas are generated at t_w from the same configuration but with independently sampled velocities. In this study, $N_s = 8-40$, and $N_r = 6-40$. The quantities $q_l(t, t_w)$ and $Q_l(t, t_w)$ are obtained by averaging over the specified replica pairs and independent samples.

Correspondingly, global correlation functions are defined. The global autocorrelation functions are defined as

$$c(t, t_w) = \frac{1}{L-1} \sum_{l=1}^{L-1} c_l(t, t_w), \quad (17)$$

$$C(t, t_w) = \frac{1}{L} \sum_{l=1}^L C_l(t, t_w). \quad (18)$$

The autocorrelation functions characterize how fast the system relaxes. The global replica-correlation functions

are,

$$q^{ab}(t, t_w) = \frac{1}{L-1} \sum_{l=1}^{L-1} q_l^{ab}(t, t_w), \quad (19)$$

$$Q^{ab}(t, t_w) = \frac{1}{L} \sum_{l=1}^L Q_l^{ab}(t, t_w). \quad (20)$$

E. Dataset

The input dataset contains the MNIST handwritten digit images [31, 32], which are widely used in machine learning benchmarking. The original images are coarse-grained into 28×28 -pixel pictures, and each pixel is either white (+1) or black (−1). We only select images of two digits, “0” and “1”, from the database for a simple task of binary classification. The input dataset is divided into a training set of 2000 images and a test set of 400 images.

TABLE I. **Parameters of M , N , and $\ln \alpha = \ln(M/N)$ used in this study.** Two settings are listed: fixed $M = 2000$ and fixed $N = 10$. At fixed $M = 2000$, the current algorithm allows simulations up to $N_{\max} = 40$.

$\ln \alpha$	M (fixed)	N	M	N (fixed)
0	-	-	10	10
0.7	-	-	20	10
1.4	-	-	40	10
2.1	-	-	80	10
2.8	-	-	160	10
3.2	-	-	250	10
3.9	2000	40	500	10
4.6	2000	20	1000	10
5.3	2000	10	2000	10
6.0	2000	5	4000	10

III. RESULTS

We systematically characterize how training dynamics gives rise to the solid–liquid–solid structure and its implications for generalization and landscape geometry.

A. Global training dynamics: aging and relaxation

We first characterize the global training dynamics of the system. Figure 3(a–d) shows the time evolution of the reduced energy $E(t)/E(0)$ and the test accuracy $A(t)$, obtained by evaluating the instantaneous weights $\{\mathbf{J}_l(t)\}$ in a feed-forward network (see Appendix A for other T).

The dynamics exhibits a clear two-step relaxation. At short times ($t < \tau_0$), the energy decreases rapidly while the test accuracy remains nearly constant, $A(t) \approx 0.5$, reflecting local relaxation without improving generalization. In the intermediate regime ($\tau_0 < t < \tau_{\text{th}}$), both en-

ergy and accuracy evolve logarithmically in time, indicating non-equilibrium aging. Such logarithmic relaxation, $E(t) \sim -\ln t$, is a hallmark of glassy dynamics, reflecting slow exploration of a rugged energy landscape, and has been observed in disordered systems [35]. Similar logarithmic decay of the loss function has also been reported in stochastic gradient descent training of standard machine learning models [11]. At long times ($t > \tau_{\text{th}}$), the system approaches a steady state where both observables saturate.

This scenario is confirmed by the waiting-time dependence of the autocorrelation function $c(t, t_w)$, as shown in Fig. 3(g). The dependence on t_w in the intermediate regime demonstrates aging, while the collapse of the curves at long times indicates effective thermalization. Consistently, correlation functions evaluated at long times become nearly independent of t_w after $\tau_{\text{th}} \approx 10^4$ (Fig. 3(h)). Other global correlation functions (C, q, Q) exhibit consistent behavior (see Fig. 4), confirming that the observed aging dynamics is a generic feature of both spins and weights.

These results indicate that the training dynamics is intrinsically non-equilibrium, characterized by aging and a well-defined thermalization time scale τ_{th} .

B. Generalization as a competition between constraints and frustration

We next examine how generalization depends on the constraint density $\alpha = M/N$. The late-time values of the energy E^* and accuracy A^* at $t^* = 5 \times 10^5$ are shown in Fig. 3(e,f).

We observe a non-monotonic dependence of $A^*(\alpha)$, with a maximum separating the overparameterized regime (small α) from the over-constrained regime (large α). This behavior reflects a competition between two effects.

On one hand, increasing α introduces more constraints, which enhances information extraction from the data and improves generalization. On the other hand, excessive constraints induce frustration, leading to a degradation of performance. This competition provides a dynamical perspective on generalization, linking performance to the balance between constraint satisfaction and configurational complexity. This interplay suggests that generalization can be understood as an emergent property controlled by a balance between constraint density and configurational freedom.

C. Layer-resolved dynamics and emergence of solid–liquid–solid structure

We now resolve the dynamics at the level of individual layers. Figure 6(a) shows the spin autocorrelation functions $c_l(t, t_w)$ for different layers.

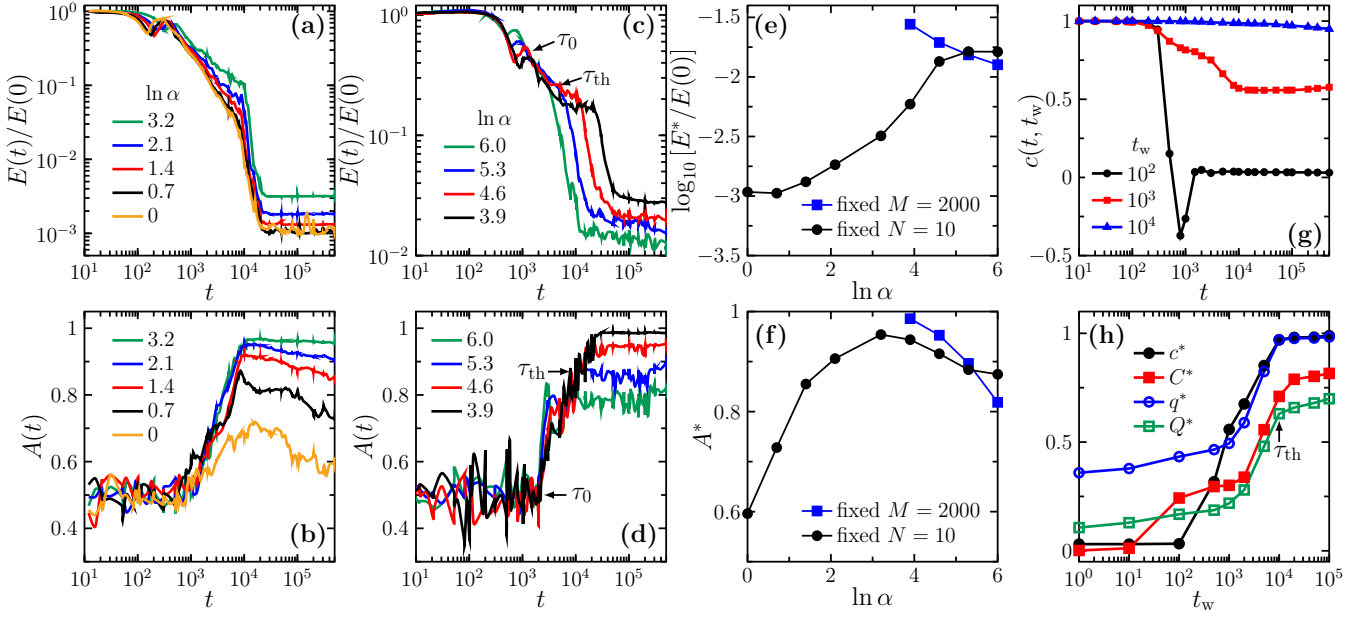


FIG. 3. **Global training dynamics and test accuracy.** Time evolution of the (a,c) reduced potential energy $E(t)/E(0)$ and (b,d) test accuracy $A(t)$, for different $\ln \alpha = \ln(M/N)$. In (a,b), $M = 2000$ is fixed and in (c,d) $N = 10$ is fixed (see Table I). (e,f) $E^* = E(t^*)$ and $A^* = A(t^*)$, where $t^* = 5 \times 10^5$ is the maximum simulation time. (g) Global autocorrelation function of spins $c(t, t_w)$. (h) The values of different correlation functions at t^* are plotted as functions of t_w . In (g,h), $\ln \alpha = 5.3$ and $M = 2000$, for which $\tau_0 \approx 10^3$ and $\tau_{th} \approx 10^4$ (marks).

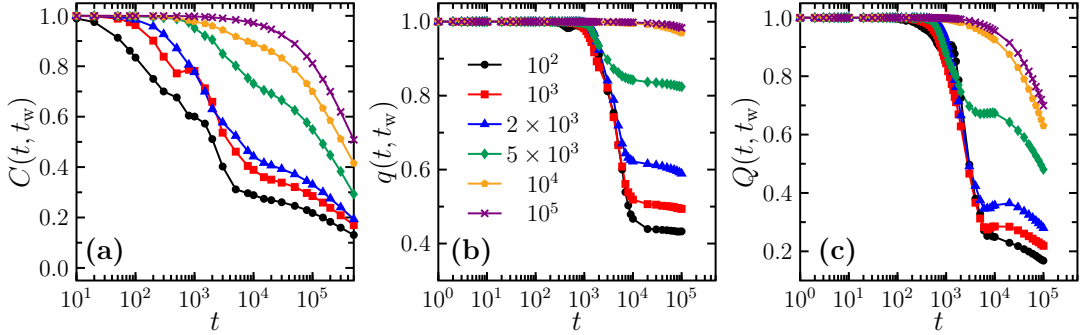


FIG. 4. **Global correlation functions.** (a) Autocorrelation function $C(t, t_w)$ of weights. (b) Replica-correlation function $q(t, t_w)$ of spins. (c) Replica-correlation function $Q(t, t_w)$ of weights. The legend indicates t_w . We set $M = 2000$ and $\ln \alpha = 5.3$.

A clear qualitative difference emerges: central layers decorrelate rapidly, while boundary layers relax much more slowly. This indicates liquid-like dynamics in the center and solid-like behavior near the boundaries.

To quantify this, we define a relaxation time τ_l^c via $c_l(t = \tau_l^c, t_w) = 1/e$. As shown in Fig. 6(b), τ_l^c reaches a plateau for central layers when $t_w > \tau_{th}$, indicating equilibration. In contrast, in boundary layers, $c_l(t, t_w)$ decays too slowly to reach the threshold $1/e$ up to the maximum time $t = 5 \times 10^5$, when t_w is large (in this case, τ_l^c cannot be estimated). In other words, boundary layers remain out of equilibrium on the same time scale.

A complementary characterization is provided by the replica overlap $q_l(t, t_w)$. As shown in Fig. 6(c,d), q_l^* remains small in central layers but finite near the

boundaries. Unlike autocorrelations, the replica overlap directly probes the similarity between independently evolved configurations, making it a suitable indicator of the underlying phase structure.

We therefore define liquid layers by $q_l^* < 1/e$ and solid layers otherwise. We have verified that the qualitative phase structure is robust against moderate variations of this threshold. The liquid and solid depths are ξ_l and $\xi_s = L - \xi_l$ respectively (see Fig. 6(d)). We use this method to quantitatively determine liquid and solid layers in the network. The results with varied $\ln \alpha$ are summarized by the numerical phase diagram Fig. 1(b) (see Appendix B for additional data).

An important qualification concerns the role of the storage ratio $\alpha = M/N$. In the replica theory for unstruc-

tured random data, the solid-liquid phase boundaries depend on α in the joint large- M , large- N scaling limit. In our simulations with structured MNIST data and finite network widths, however, M and N have non-equivalent effects. As shown in Fig. 5, varying N at fixed $M = 2000$ produces a substantially stronger change in the layer-resolved overlaps than varying M at fixed $N = 10$. This difference may arise from both finite-width effects and correlations in the MNIST data, for which the effective number of independent constraints, M_{eff} , can be smaller than the nominal number of training samples M .

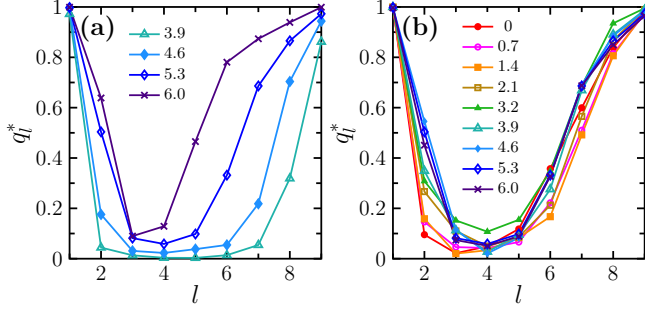


FIG. 5. **Solid-liquid-solid network structure for different storage ratio α .** We plot $q_l^* \equiv q_l(t = 5 \times 10^5, t_w = 5 \times 10^4)$ as a function of l , for a few different $\ln \alpha = \ln(M/N)$. (a) $M = 2000$ fixed and N varied. (b) $N = 10$ fixed and M varied.

The quantities q_l^* and Q_l^* are finite-time observables evaluated at the largest accessible simulation time. Therefore, the numerical phase boundary in Fig. 1(b) should be interpreted as an operational finite-time crossover boundary. An additional long-time test for a smaller network is presented in Appendix C, confirming that the qualitative layer-dependent structure remains robust when the overlap functions have evolved to their asymptotic plateaus.

To test whether the layer-dependent liquid-like behavior is merely a consequence of the permutation symmetry of the fully connected architecture, we additionally study a quenched fixed-degree sparse network with width $N = 5$ and connectivity $K = 3$ ($p = K/N = 0.6$; see Appendix D). Both the autocorrelation functions and the replica-overlap functions retain a pronounced layer dependence, with smaller correlations in the intermediate layers than near the boundaries, supporting the interpretation that the observed heterogeneous dynamics is robust against random dilution.

These results demonstrate the emergence of a solid-liquid-solid structure along the depth of the network, consistent with theoretical predictions. The layer-dependent structure has direct implications for learning. In the central liquid layers, configurations decorrelate across replicas, providing a large space of accessible states and thus strong representation capability. In contrast, boundary layers exhibit strong correlations and slow dynamics, reflecting the constraints imposed by the input

and output data. These layers stabilize the learned mapping and support generalization. This coexistence naturally realizes a separation between flexible representation in the bulk and constraint enforcement at the boundaries. This observation suggests a physical origin of generalization in terms of a spatial separation of roles across layers. We further observe that beyond a critical depth, the thickness of the solid layers saturates, while additional depth only extends the liquid region (see Appendix E), indicating that the structure is controlled by boundary constraints rather than bulk properties.

D. Structure of the solution space: liquid vs solid layers

To probe the structure of the solution space, we analyze the overlap order parameter $Q_l^{ab} \sim \mathbf{J}_l^a \cdot \mathbf{J}_l^b$, which measures the similarity between configurations a and b . In this study, Q_l^{ab} is estimated from the dynamical data, $Q_l^{ab} \equiv Q_l^{ab}(t = 10^5, t_w = 10^3)$. The distribution $P(Q_l^{ab})$ is computed from $(N_s \times N_r) \times (N_s \times N_r)$ pairs of configurations $(a, b = 1, 2, \dots, N_s \times N_r)$, with N_s independent random initial configurations at $t = 0$ (samples), and N_r replicas for each initial configuration replicated at t_w (see Appendix F for additional data). The matrix element Q_l^{ab} is sorted using the complete-linkage (CL) agglomerative clustering algorithm [36] (see Appendix G for details). This method can identify hierarchical groups of configurations with similar overlaps. Based on the clustered groups, the states are organized into a tree structure. The configurations in the lowest sub-trees have the largest overlaps Q_l^{ab} , and configurations with smaller overlaps are grouped in higher-level sub-trees. The procedure is performed recursively in the descending order of Q_l^{ab} .

In a typical liquid layer ($l = 5$), the states are well-connected and the solution space is structureless. This property is reflected by a Gaussian distribution $P(Q_l^{ab})$ with a zero mean, a random matrix, and a single-level tree (Fig. 7(a-c)).

The solution space structure in a solid layer ($l = 9$) is more complex (Fig. 7(d-f)). Crystal-like solid layers are essential for generalization [14, 15], while glassy states may emerge in these solid layers due to frustration caused by a large number of constraints. This expectation is consistent with our numerical results. The $P(Q_l^{ab})$ in Fig. 7(d) has two peaks: the right one corresponds to the overlaps between configurations replicated from the same sample (same solid), and the left one corresponds to those in different samples (different solids). Multiple solid states appear due to the permutation symmetry in the fully connected DNN. What is more fascinating is that, within a solid state picked up by a given sample, delicate hierarchical structures can be identified. We zoom in on one typical sample by increasing the number of replicas N_r for the given sample. As shown in Fig. 7(f), the diagonal block in the matrix of Fig. 7(e) actually

contains hierarchical sub-blocks, and correspondingly the sub-tree has hierarchical sub-sub-trees. Thus the solution space in a crystal state is not structureless, but instead consists of hierarchical clusters of solutions as suggested by the mean-field calculations [14]. A similar picture has been proposed for the phase space of spin and structural glasses in the context of the *Gardner transition* [37–40], where hierarchical structures are found in multiple glass (instead of crystal) states. The transition from liquid to glass layers is further reflected in the distribution $P(Q_l^{ab})$, which evolves from Gaussian to highly non-trivial forms near the boundaries (see Appendix F).

These observations allow us to infer the underlying structure of the learning landscape. We find that the solution space in solid layers exhibits a hierarchical organization, while remaining structureless in liquid layers. This layer-dependent organization provides a direct dynamical signature of the underlying landscape structure. It suggests that the landscape explored during training is fundamentally heterogeneous, with qualitatively different structures coexisting within a single network. This provides a direct bridge between dynamical observations and the underlying landscape geometry, linking training dynamics to solution space organization. This suggests that the learning process does not explore a homogeneous landscape, but rather navigates between regions with fundamentally different geometrical structures.

IV. DISCUSSION

Our results suggest a coherent dynamical and structural picture of deep learning that directly connects training dynamics, layer organization, landscape geometry, and generalization. The coexistence of liquid-like central layers and solid-like boundary layers leads to a natural separation between flexible representations and constrained outputs.

Building on the connection between dynamics and landscape geometry, we further interpret the hierarchical organization in terms of glassy physics. Our results suggest that generalization is not solely determined by optimization or regularization, but emerges from a spatial organization of degrees of freedom across layers. While the replica theory provides a static description of the solution space, our results reveal how such structures emerge dynamically during training. In particular, the observed aging and timescale separation highlight the importance of non-equilibrium processes beyond mean-field descriptions.

The solution space in solid layers exhibits a hierarchical organization, characterized by nested clusters of solutions. This structure is reminiscent of glassy systems and suggests that the landscape is organized into multiple metastable basins. The hierarchical organization observed in solid layers suggests a direct connection to Gardner-like marginal stability, indicating that deep learning systems may operate near a critical regime. This

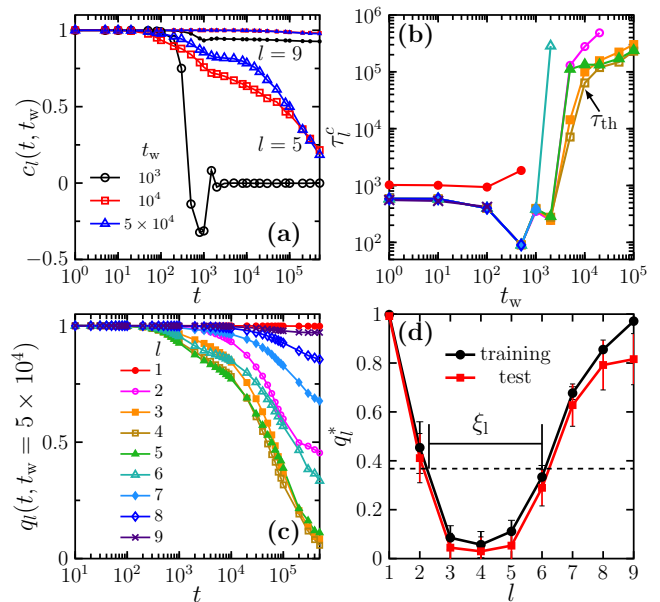


FIG. 6. **Layer-dependent training dynamics.** (a) Spin autocorrelation functions $c_l(t, t_w)$ in the $l = 5$ (open symbols) and $l = 9$ (filled symbols) layers, for $t_w = 10^3$ (black), 10^4 (red), 5×10^4 (blue). (b) Spin correlation time τ_l^c as a function of t_w , with $\tau_{th} \approx 10^4$ indicated, for different l (colors have the same meaning as in (c)). (c) Spin replica-correlation functions $q_l(t, t_w = 5 \times 10^4)$ in different layers. (d) $q_l^* \equiv q_l(t = 5 \times 10^5, t_w = 5 \times 10^4)$ as a function of l . For comparison, the spin overlaps of the internal representations generated during testing are also plotted. We fix $\ln \alpha = 5.3$ and $M = 2000$ in all panels. Error bars represent standard errors.

provides a possible microscopic origin of the observed separation between flexibility and stability across layers.

The present system size is limited by the $\mathcal{O}(LN^2)$ weight variables and $\mathcal{O}(LMN)$ internal-representation variables evolved in the MD simulations. Increasing the number of output constraints is not expected to change the qualitative scaling of the solid-layer penetration depth predicted by the replica theory [14, 15], although it may affect the prefactor and finite-size crossover.

The observed phenomena are not specific to MD training. Similar slow relaxation and aging behaviors have been reported in stochastic gradient descent, suggesting that these features reflect generic properties of high-dimensional optimization landscapes. This perspective may provide new insights into the interplay between representation, generalization, and training dynamics in modern deep learning systems. These features are expected to be generic to overparameterized systems with boundary constraints, beyond the specific TDLM framework. In addition, we have checked that standard Monte Carlo (MC) simulations of the same model on Ising spins reproduce the same qualitative features, including aging behavior and layer-dependent relaxation, further supporting that these phenomena are robust with respect to the choice of dynamical implementation. However, stan-

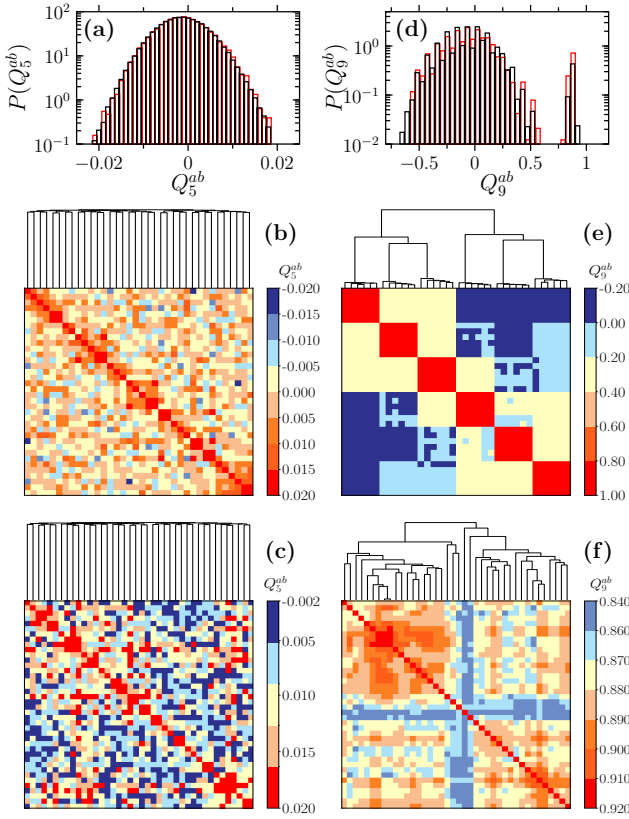


FIG. 7. Structure of the solution space in typical liquid and solid layers revealed by overlap order parameters. Data in (a–c) and (d–f) are respectively for typical liquid ($l = 5$) and solid ($l = 9$) layers. We set $M = 2000$ and $\ln \alpha = 4.6$. (a,d) Probability distribution $P(Q_l^{ab})$, for $(N_s, N_r) = (20, 8)$ (red) and $(38, 16)$ (black). (b,e) The 36×36 overlap matrix of Q_l^{ab} obtained with $(N_s, N_r) = (6, 6)$, where the color bar represents the value of the matrix element. (c,f) A close-up of the second sample in (b,e) by increasing the number of replicas from $N_r = 6$ to $N_r = 40$ for this sample.

dard MC dynamics are much slower than the current GPU-accelerated MD dynamics. More sophisticated MC algorithms are required to achieve the same level of relaxed total energy.

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Appendix A: Dependence of the training energy and test accuracy on the temperature

Figure 8 compares the training energy (loss function) $E(t)/E(0)$ and test accuracy $A(t)$ at different temperatures T . In all cases, the initial configurations at $t = 0$ are random ($T = \infty$), and the system is rapidly quenched to the target temperature T . Except for the highest $T = 10^{-2}$, the asymptotic test accuracy in the long-time limit is nearly independent of T (Fig. 8(b)).

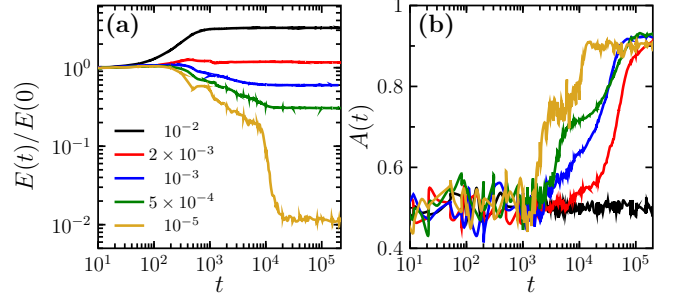


FIG. 8. Training energy $E(t)/E(0)$ and test accuracy $A(t)$ at a few different temperatures T . We set $M = 2000$ and $\ln \alpha = 4.6$.

Appendix B: Additional results on layer-specific correlation functions

In Fig. 9, we present additional data on the layer-specific correlation functions. The waiting time t_w -dependence of the weight relaxation time τ_l^C is shown in Fig. 10, where τ_l^C is defined by $C_l(t = \tau_l^C, t_w) = 1/e$ (see Fig. 6(b) for corresponding data on τ_l^C). The data, together with Figs. 5 and 6, consistently suggest a solid-liquid-solid structure of parameter states in the network.

Appendix C: Long-time convergence test in a smaller network

To further examine the finite-time nature of the layer-resolved overlap functions, we perform additional simulations for a smaller network, for which longer training trajectories are numerically more accessible. In this test, we use a network with $M = 420$, $N = 3$, and $L = 10$, and extend the MD training up to 10^6 steps. To improve the statistical quality, we consider 4 independent samples. For each sample, 5 replicas are generated at the waiting time t_w , yielding 10 replica-pair overlap values per sample. The layer-resolved overlap functions are then averaged over all replica pairs and over all four samples.

Figure 11 shows the resulting weight-overlap functions for the smaller system. Compared with the larger-system data shown in the main text, the longer simulation of the smaller network provides an additional convergence

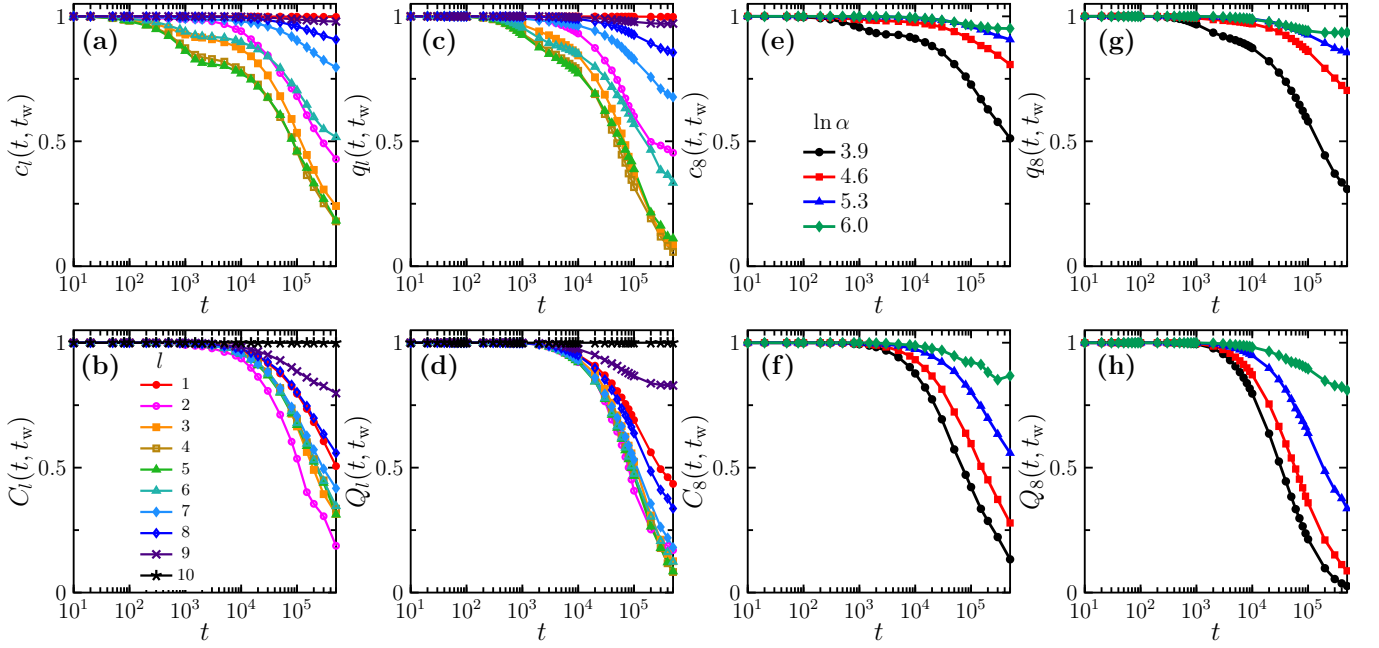


FIG. 9. **Layer-specific correlation functions.** Data are obtained with a fixed waiting time $t_w = 5 \times 10^4$. (a) Spin autocorrelation functions $c_l(t, t_w)$, (b) weight autocorrelation functions $C_l(t, t_w)$, (c) spin replica-correlation functions $q_l(t, t_w)$, and (d) weight replica-correlation functions $Q_l(t, t_w)$, for different l , with a fixed $\ln \alpha = 5.3$. (e-h) Correlation functions for the 8th layer, with a few different $\ln \alpha$.

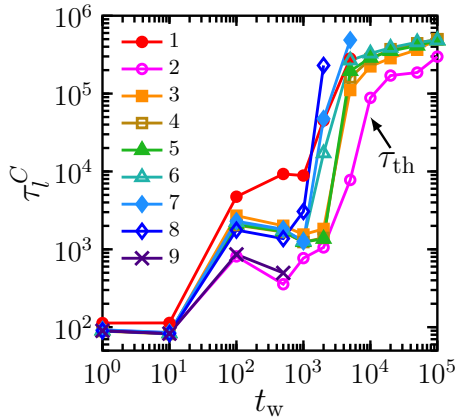


FIG. 10. **Waiting time t_w -dependence of the weight relaxation time τ_l^C .** The legend indicates l . We set $\ln \alpha = 5.3$.

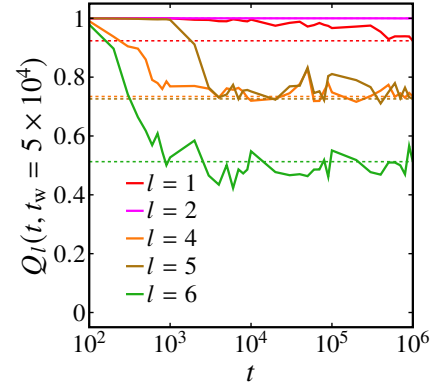


FIG. 11. **Long-time convergence of overlap functions in a smaller network.** The plot shows $Q_l(t)$ versus time t with $t_w = 5 \times 10^4$, for a few selected layers in the center and near the boundaries.

Appendix D: Randomly diluted network

check. In particular, the weight-overlap functions for the central layers have clearly converged to long-time plateaus. The results confirm that the qualitative layer-dependent structure remains robust: the internal layers remain less correlated than the layers near the boundaries, consistent with the solid-liquid-solid interpretation.

To examine whether the layer-dependent liquid-like behavior is a consequence of the permutation symmetry of the fully connected architecture, we perform additional MD simulations for a sparsely connected network.

The number of neurons in each hidden layer is set to $N = 5$, and each hidden neuron is connected to exactly $K = 3$ neurons in each adjacent hidden layer. The cor-

responding connection density is therefore

$$p = \frac{K}{N} = 0.6. \quad (\text{D1})$$

Only the hidden-to-hidden connections are diluted. The connections between the input layer and the first hidden layer, as well as those between the last hidden layer and the output layer, remain fully connected. The number of training samples is fixed at $M = 1000$, and the remaining simulation parameters, including the temperature, friction coefficient, time step, and parameter bounds, are chosen to be the same as in the fully connected simulations.

The sparse connectivity is implemented using a fixed-degree topology. We first constructed a K -regular bipartite connectivity pattern between two adjacent hidden layers and then independently relabeled the input and output nodes using random permutations. Consequently, every hidden neuron has exactly K incoming and K outgoing connections, and no hidden neuron is isolated. The resulting topology is therefore a quenched fixed-degree sparse network.

The sparse connectivity mask is generated once with a fixed random seed and kept unchanged throughout the MD evolution. The same quenched mask is used for all independent initial conditions and all replicas. Therefore, differences between samples and replicas originate from their dynamical trajectories rather than from different network topologies.

The masked weight variables are fixed to zero throughout the simulation. Their corresponding velocities are also set to zero, and these inactive degrees of freedom are excluded from the calculation of the kinetic energy and correlation functions. Figure 12 shows the resulting overlap functions. The overlap functions show a layer-dependent structure, qualitatively consistent with Fig. 6(d).

Appendix E: Dependence on the total network depth

In Fig. 13, the effects of the total network depth L are examined. We observe that when the network is deep enough, i.e., when $L \geq 10$ (for $\ln \alpha = 5.3$), the depth ξ_s of the solid layers no longer changes even when L is further increased. In such deep networks, increasing L only introduces more liquid layers.

Appendix F: Additional data on overlap distribution

$$P(Q_l^{ab})$$

In Figs. 14, 15 and 16, we plot the distributions $P(Q_l^{ab})$ in different layers ($l = 1 - 10$), for $\ln \alpha = 0, 3.2, 6.0$ (for a fixed $N = 10$). (i) When $\ln \alpha \leq 3.2$, $P(Q_l^{ab})$ is close to a Gaussian distribution in all layers. The center of the Gaussian distribution varies with l in a manner similar to the layer dependence of q_l^* shown in Fig. 6(d). (ii) When

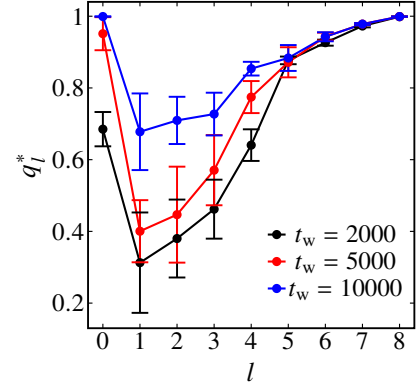


FIG. 12. **Layer-resolved overlap functions in the quenched fixed-degree sparse network.** The plot shows the spin overlaps q_l^* , for $t_w = 2 \times 10^3$, 5×10^3 , and 10^4 . For each waiting time, the data are averaged over 4 independent samples; each sample contains 5 replicas, giving 10 replica-pair overlap values per sample. The smaller overlaps in the intermediate layers compared with those near the boundaries show that the layer-dependent separation between more weakly and more strongly correlated sectors persists after sparse dilution. For all waiting times, the overlap functions are evaluated at the same elapsed time, $t^* = 5 \times 10^5$. Error bars denote standard errors.

In $\ln \alpha > 3.2$, only the liquid layers still have a Gaussian-like $P(Q_l^{ab})$. Approaching the input or output boundary, a non-trivial distribution gradually emerges, suggesting a hierarchical evolution of the solution space from a simple structure in liquid layers to a complex structure in solid layers [14].

Appendix G: Clustering algorithm to construct the overlap matrix and the corresponding tree

We employ the CL clustering algorithm [36] to generate the heat map of the overlap parameter matrix and the corresponding tree in Fig. 7 in the main text. The method groups data points into hierarchical clusters based on their maximum pairwise distances. We define the distance between every pair of replicas in the l -th layer, using a metric $d_l^{ab} \equiv 1 - Q_l^{ab}$. The CL intercluster distance (dissimilarity) is the largest distance between any member a from the first cluster A (i.e., $a \in A$) and any member b from the second cluster B (i.e., $b \in B$), $D_l^{AB} = \max_{a \in A, b \in B} d_l^{ab}$.

The clustering process is outlined as follows.

i) *Generating initial clusters.* Consider each replica as an individual cluster.

ii) *Finding the closest clusters.* Find a pair of clusters A and B with the minimum CL intercluster distance D_l^{AB} .

iii) *Merging clusters.* Merge the two closest clusters A and B into C . Update the CL intercluster distances between C and other clusters. Continue merging the closest clusters until all data points are consolidated into a single

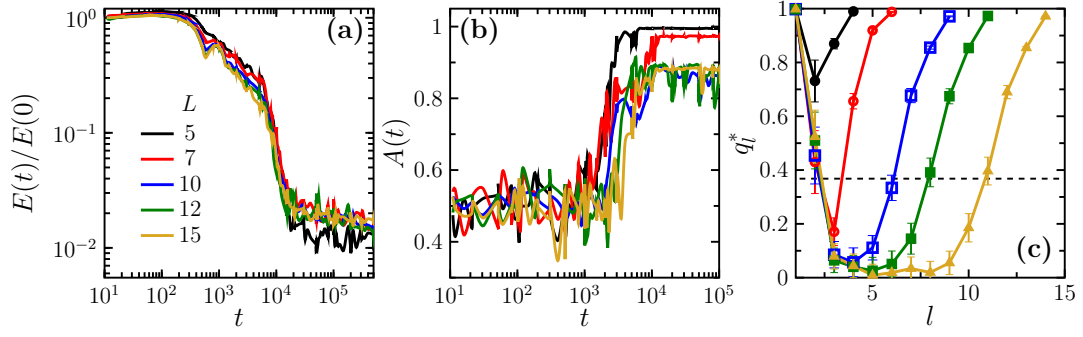


FIG. 13. **Effects of the total network depth L .** (a) Training energy density $E(t)/E(0)$ and (b) test accuracy $A(t)$ for different L ($\ln \alpha = 5.3$). (c) Spin replica-correlation $q_l^* = q_l(t = 5 \times 10^5, t_w = 5 \times 10^4)$ as a function of l , for different L . The dashed horizontal line corresponds to $1/e$.

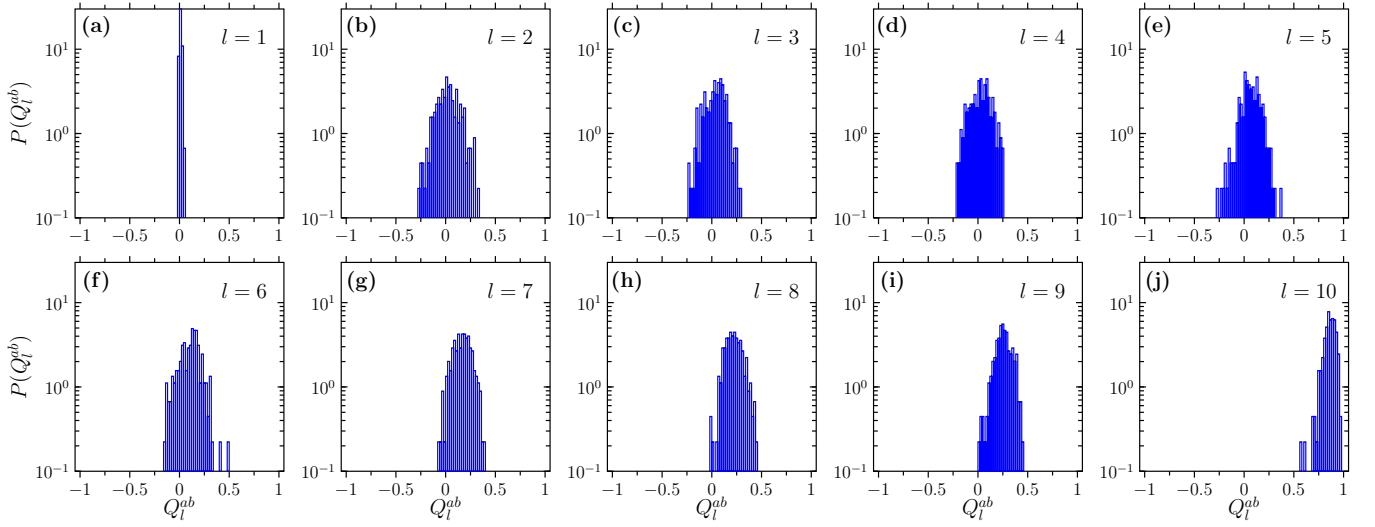


FIG. 14. $P(Q_l^{ab})$ in different layers. We define $Q_l^{ab} \equiv Q_l^{ab}(t = 5 \times 10^5, t_w = 10^3)$, and set $\ln \alpha = 0$.

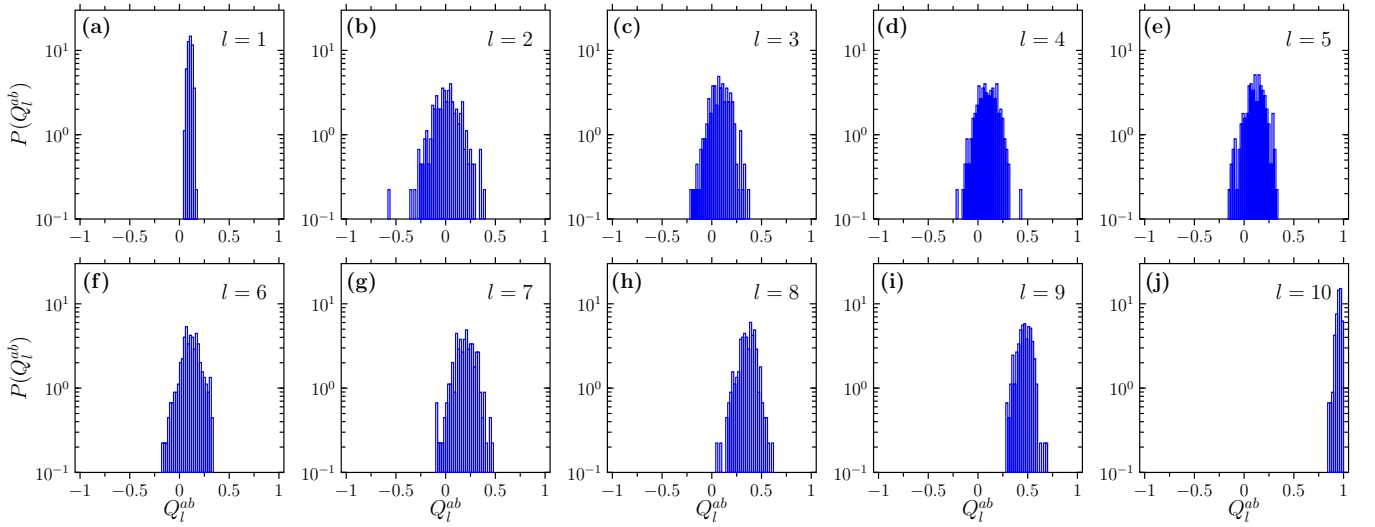


FIG. 15. $P(Q_l^{ab})$ in different layers. We define $Q_l^{ab} \equiv Q_l^{ab}(t = 5 \times 10^5, t_w = 10^3)$, and set $\ln \alpha = 3.2$.

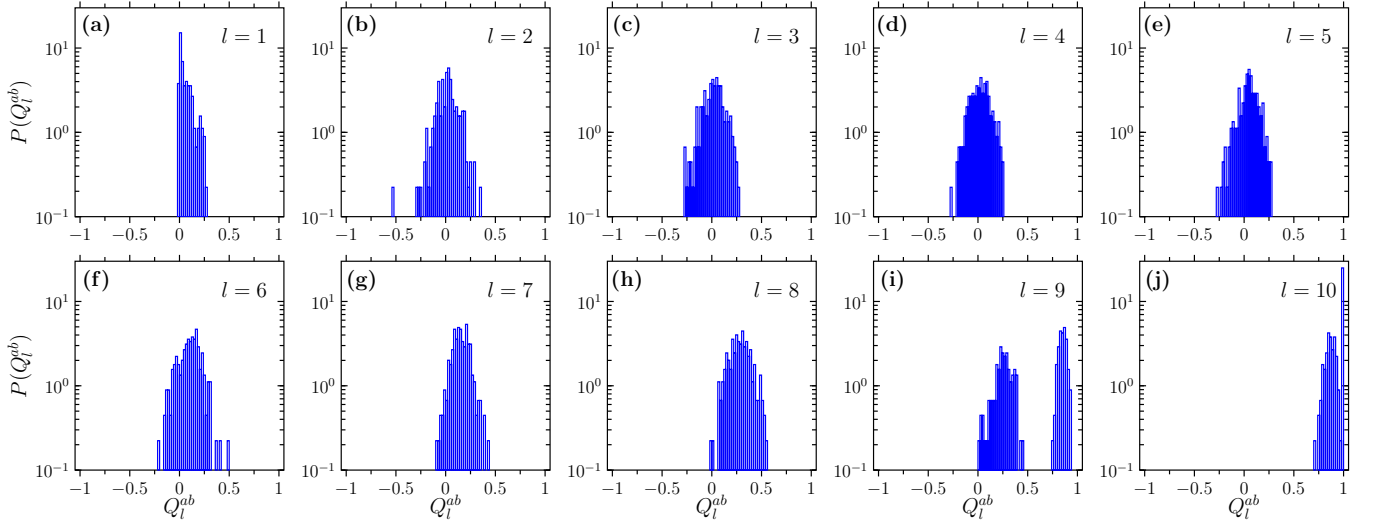


FIG. 16. $P(Q_l^{ab})$ in different layers. We define $Q_l^{ab} \equiv Q_l^{ab}(t = 5 \times 10^5, t_w = 10^3)$, and set $\ln \alpha = 6.0$.

cluster.

iv) Constructing a dendrogram. Construct a dendrogram that records the sequence of merging and the corresponding CL intercluster distances. This sequence can be visualized by a tree as shown in Fig. 7 in the main

text.

The above algorithm is implemented by using the `dendrogram` and `linkage` functions in the `scipy.cluster.hierarchy` module of the SciPy program [41].

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